DESIGN OF EXPERIMENTS × GPU KERNELS

Hardware-aware autotuning of a CUDA GEMM kernel with *response surface methodology*

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HARDWARE
NVIDIA H200 · CUDA 13.0

Same code, six numbers, *2.6×* apart in speed

UNTUNED KERNEL · TFLOPS

13.2

textbook tiles
`(64, 64, 8, 4, 4, 1)`

TUNED BY DOE-RSM · TFLOPS

34.3

selected configuration
`(128, 128, 32, 8, 8, 8)`

LEGAL CONFIGURATIONS

1,396

of 2,304; each needs its own compile and benchmark

Exhaustive search is slow and grows with every new factor; machine-learning autotuners need many samples. **Can classical statistics find a near-best configuration from a small sample?**

Five questions, each answered by measurement

RQ1	SCREENING	Can a fractional factorial design separate the parameters that matter from noise?
RQ2	MODEL	Does a second-order response surface capture the GPU’s non-linear behaviour?
RQ3	EFFICIENCY	With the same number of trials, does DOE–RSM beat random search?
RQ4	OPTIMALITY	How close does it get to the exhaustive-search optimum?
RQ5	CONTEXT	What share of cuBLAS throughput does the tuned kernel reach?

- The main claim is **RQ3, sample efficiency**. RQ5 is context, not the goal.

A four-stage pipeline, judged against three baselines

01

Screen

2^{6-1} fractional factorial · 32 configs · ANOVA

→

02

Model

central composite design · 48 points · quadratic OLS

→

03

Optimise

constrained SQP + enumeration of the fitted model

→

04

Validate

15 candidates measured · fastest is chosen

JUDGED AGAINST

Exhaustive search

all 1,396 configurations: the true optimum

Random search

same budget, repeated 10,000 times

cuBLAS

NVIDIA’s library, FP32, interleaved

Six compile-time factors, fenced in by the hardware

FACTOR	MEANING	LEVELS
BM	block tile rows · shared memory	32 64 128 256
BN	block tile columns · shared memory	32 64 128 256
BK	K-depth of each tile	8 16 32 64
TM	rows per thread · registers	2 4 8
TN	columns per thread · registers	2 4 8
UNROLL	inner-loop unroll factor	1 2 4 8

FEASIBLE

1,396

of 2,304 combinations

over 1,024 threads · **528**

over 48 KiB shared memory · **168**

tile-load overflow · **140**

fractional load stride · **72**

Filter checked against the compiler on all 2,304: no disagreements.

Rules fixed in advance; every number checked

CLOCK LOCKED

1830

MHz on all 2,001 measurements; block-to-block variation 0.033%

RANDOMISED BLOCKS

1.86

Durbin–Watson on run order: no drift (a fixed-order pilot gave 0.60)

SHARED GPU

37

measurements overlapped other users’ jobs, were detected and re-run

EVERY OUTPUT CHECKED

0

configurations disagree with cuBLAS; a planted bug was caught at 0.044

All six factors matter; *unrolling* matters most

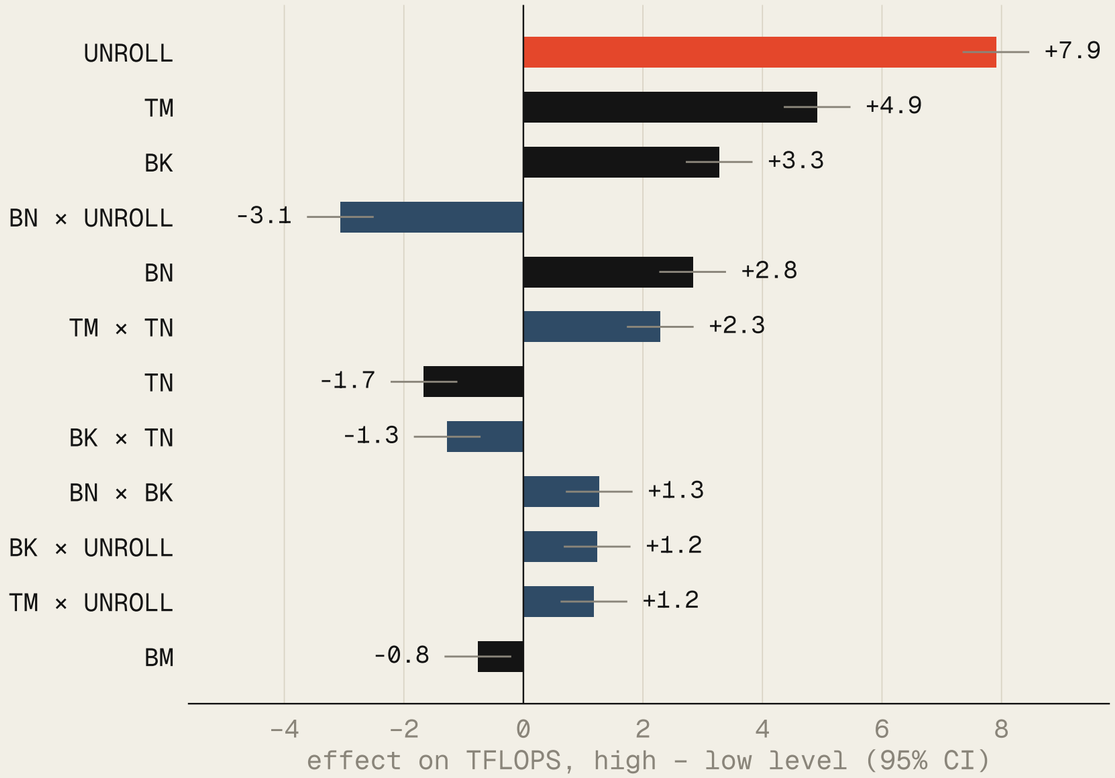


FIG. 1 — EFFECTS AT 4096^3, 2^{6-1} DESIGN, 3 RANDOMISED BLOCKS

MODEL FIT

R^2 0.959

32 configurations × 3 blocks; replicate noise 0.04%

LARGEST EFFECT

+7.9

TFLOPS from unrolling 1 → 8; 54% of all variance

RETAINED

6 of 6

by the pre-set rule $p < 0.05$; weakest BM at $p = 0.0026$

What matters changes for *small* matrices

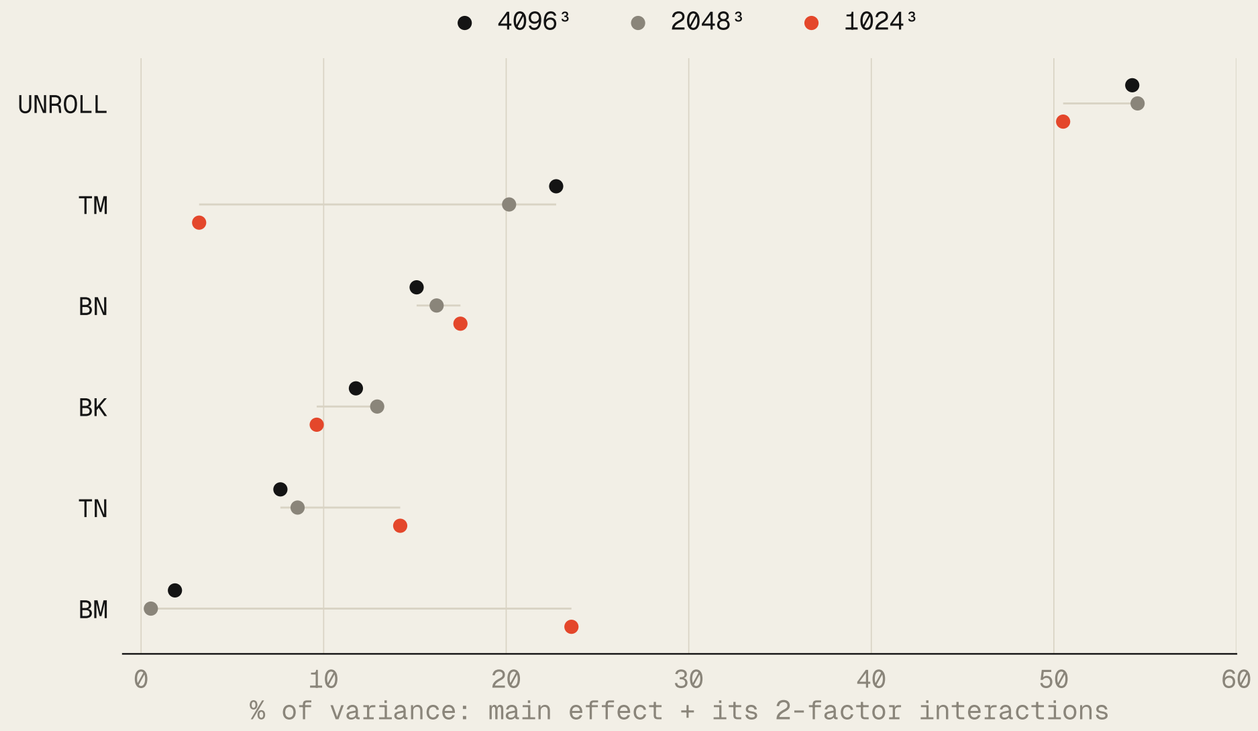


FIG. 2 — SHARE OF VARIANCE PER FACTOR AT THREE MATRIX SIZES

4096³ VS 2048³

$\tau = 1.0$

identical factor ranking

4096³ VS 1024³

$\tau = -0.07$

BM rises 1.9% → 23.6%; TM falls 22.7% → 3.2%

LIKELY CAUSE

64 / 132

thread blocks vs SMs with 128 × 128 tiles at 1024³: tile size sets GPU fill

A quadratic surface is close, but *statistically inadequate*

MODEL · 6-FACTOR CCD · 144 RUNS	R^2	LACK OF FIT	GATE · LIMIT 5%	HELD-OUT ERROR
Quadratic · pre-registered	0.956	7.8%	fails	10.6%
Quadratic on log latency	0.961	7.0%	fails	11.4%
Quadratic + 3-way terms	0.982	10.1%	fails	32.6%

Replicate noise is about 0.05%, so even small systematic misfit is detectable. The fitted quadratic is a saddle with no interior maximum.

Three-way interactions and compiler *jumps*

3-WAY INTERACTIONS

6 of 10

estimable pairs significant; strongest BM·BN·BK = TM·TN·UNROLL, $p = 10^{-13}$

ADDING THEM

10.6 → 32.6%

held-out error: explains 59% of the misfit but overfits, with 40 parameters on 45 configurations

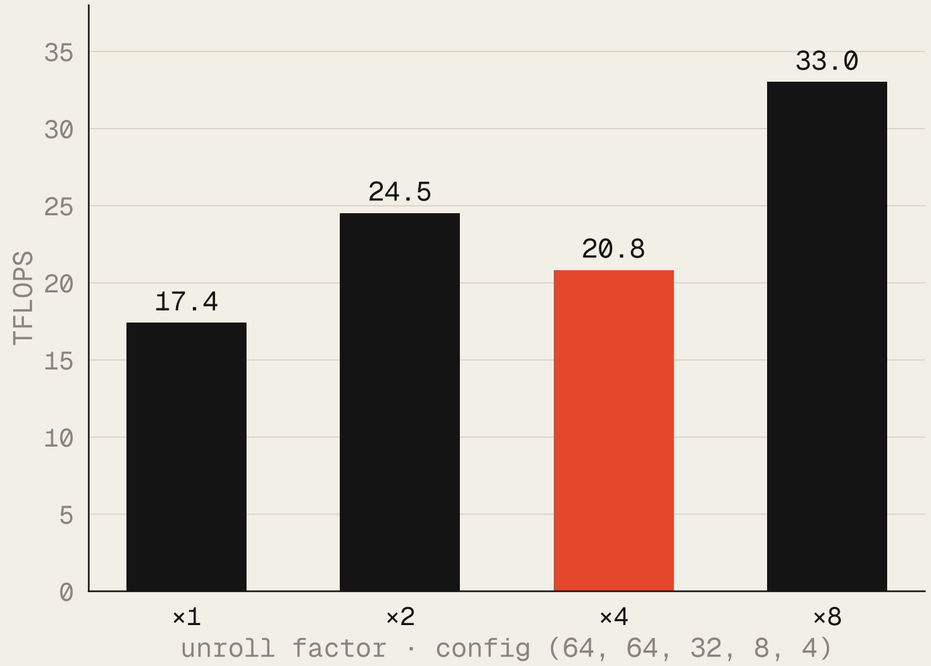
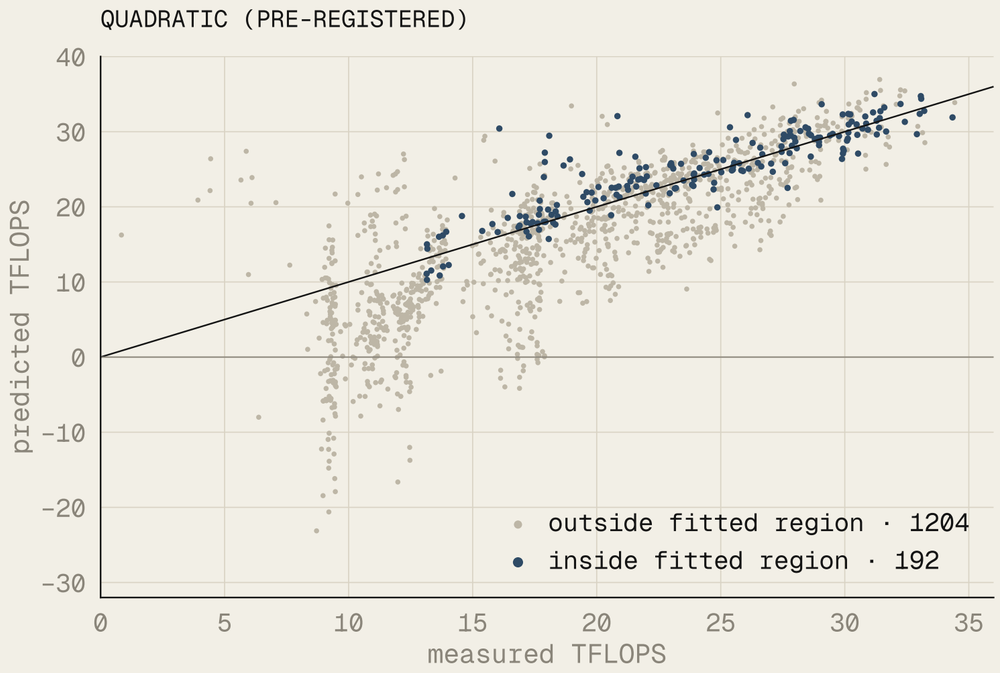
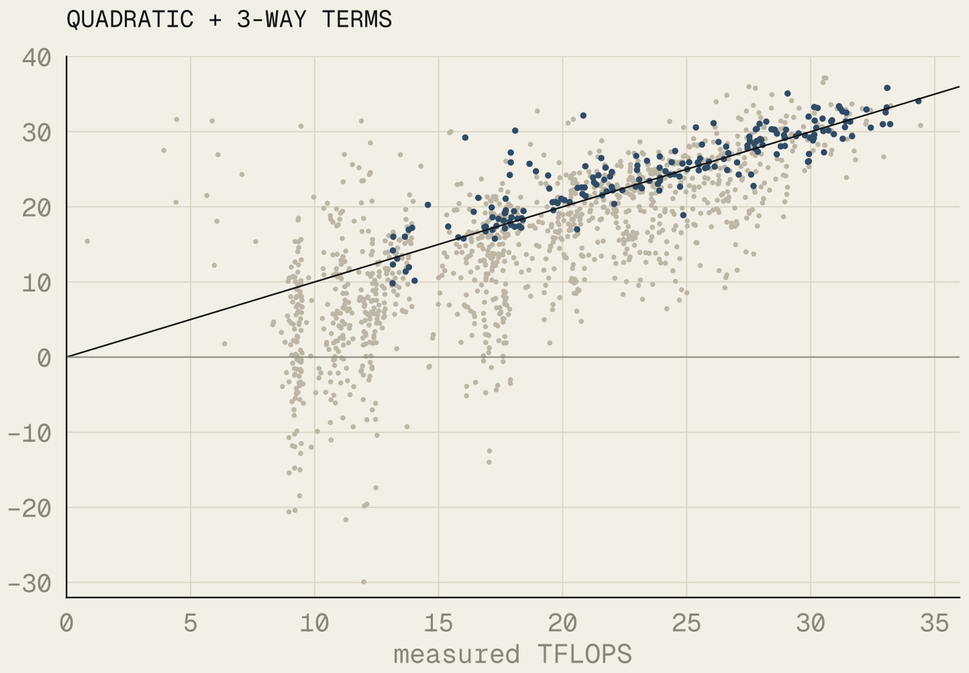


FIG. 3 — UNROLLING BY 4 IS SLOWER THAN BY 2

Trustworthy only inside the region it was fitted on





INSIDE FITTED REGION

≈ 5% median error

OUTSIDE

≈ 22%, some predictions negative

BEST GUESS OUTSIDE

37.1 predicted, 30.5 measured

Measurement, not the model, *picked the winner*

SQP and enumeration proposed 15 candidates from both models; each was measured in 5 randomised blocks.

CONFIGURATION	MEASURED TFLOPS	MEASURED RANK	AUGMENTED RANK	QUADRATIC RANK
(128, 128, 32, 8, 8, 8) · selected	34.32	1	3	11
(128, 128, 32, 8, 8, 4) · augmented #1	33.08	2	1	3
(64, 128, 32, 8, 4, 4) · quadratic #1	31.17	8	8	1

Trusting either model's own first choice would have cost **3.6%** (augmented) or **9.2%** (quadratic).

59 evaluations, *0.17%* from the optimum

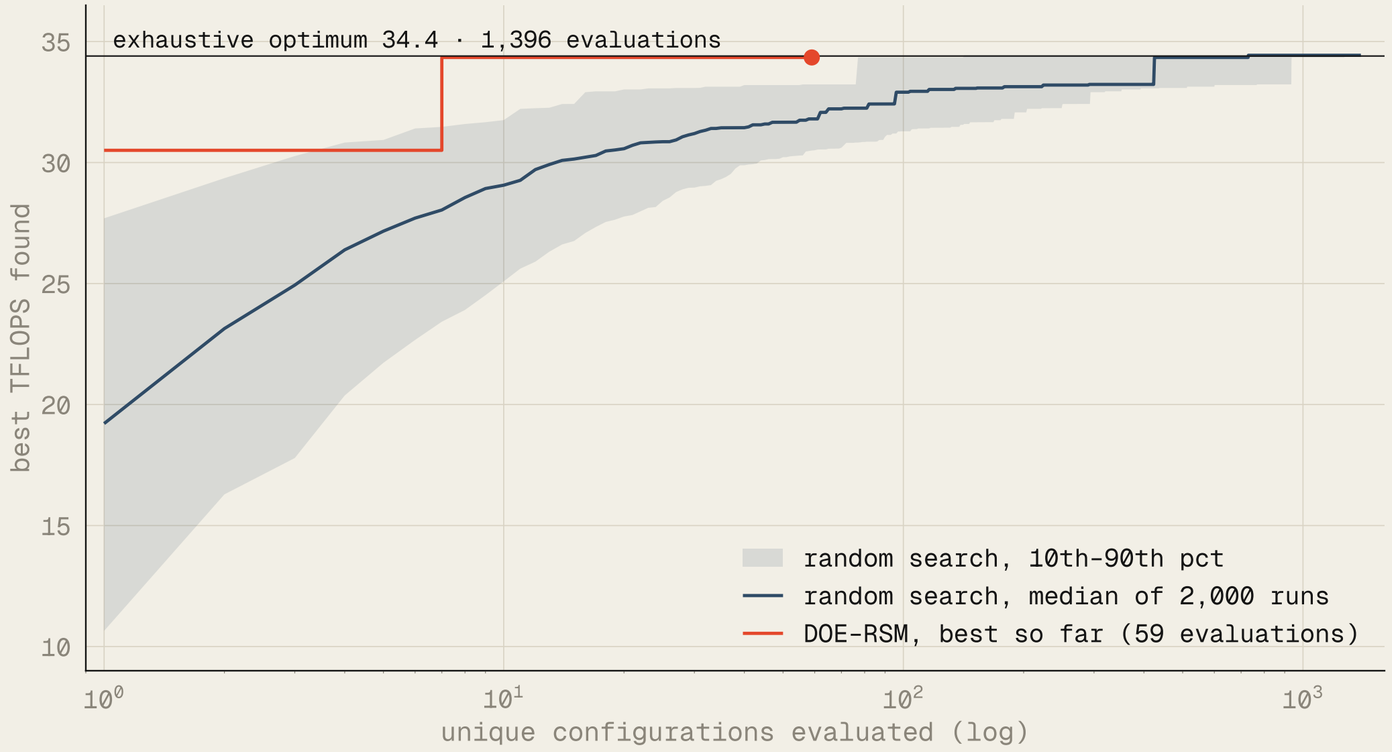


FIG. 4 — BEST CONFIGURATION FOUND VS EVALUATIONS, 4096^3

RANK

#2 of 1,396

0.17% slower than the exhaustive optimum, 34.40 TFLOPS

RANDOM SEARCH

424

median evaluations to match it: about 7× the budget

SAME BUDGET

7.8%

of 10,000 random trials did as well

72% of cuBLAS, *2.6×* the untuned kernel

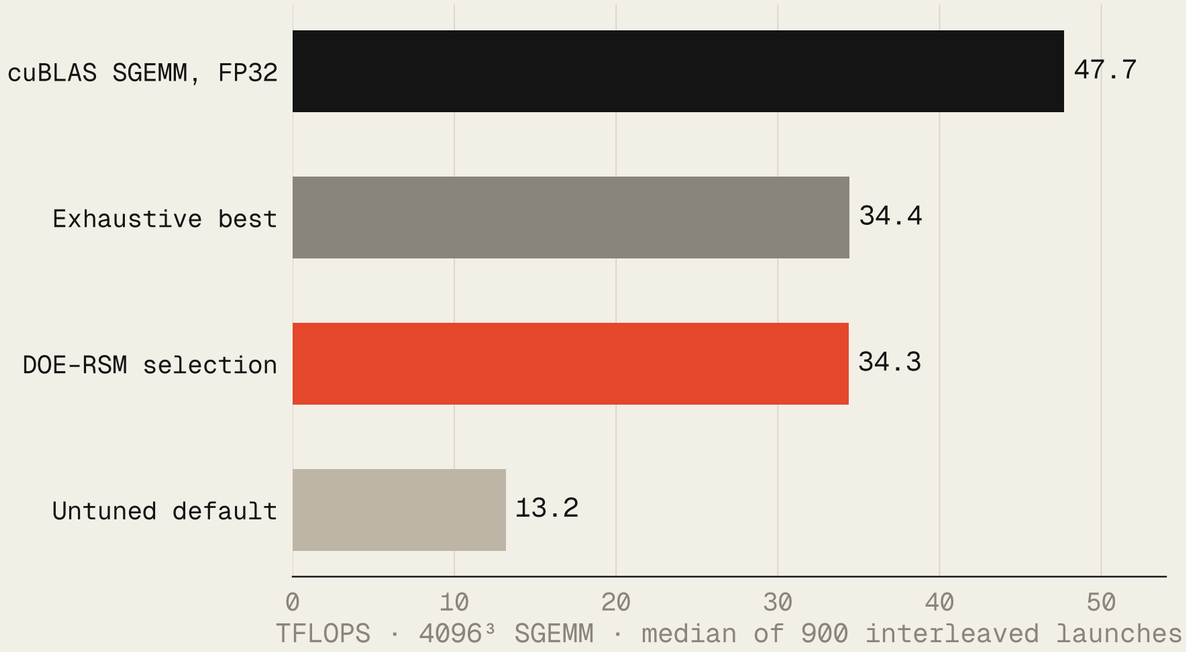


FIG. 5 — MEDIAN OF 900 INTERLEAVED LAUNCHES EACH, PEDANTIC FP32

SHARE OF CUBLAS

72.0%

34.35 vs 47.70 TFLOPS

LIKE FOR LIKE

FP32

TF32 off; outputs identical bit for bit

CEILING

72.2%

even the best of all 1,396 configurations: the gap is the kernel design

Register spilling is rare here and *not* what limits speed

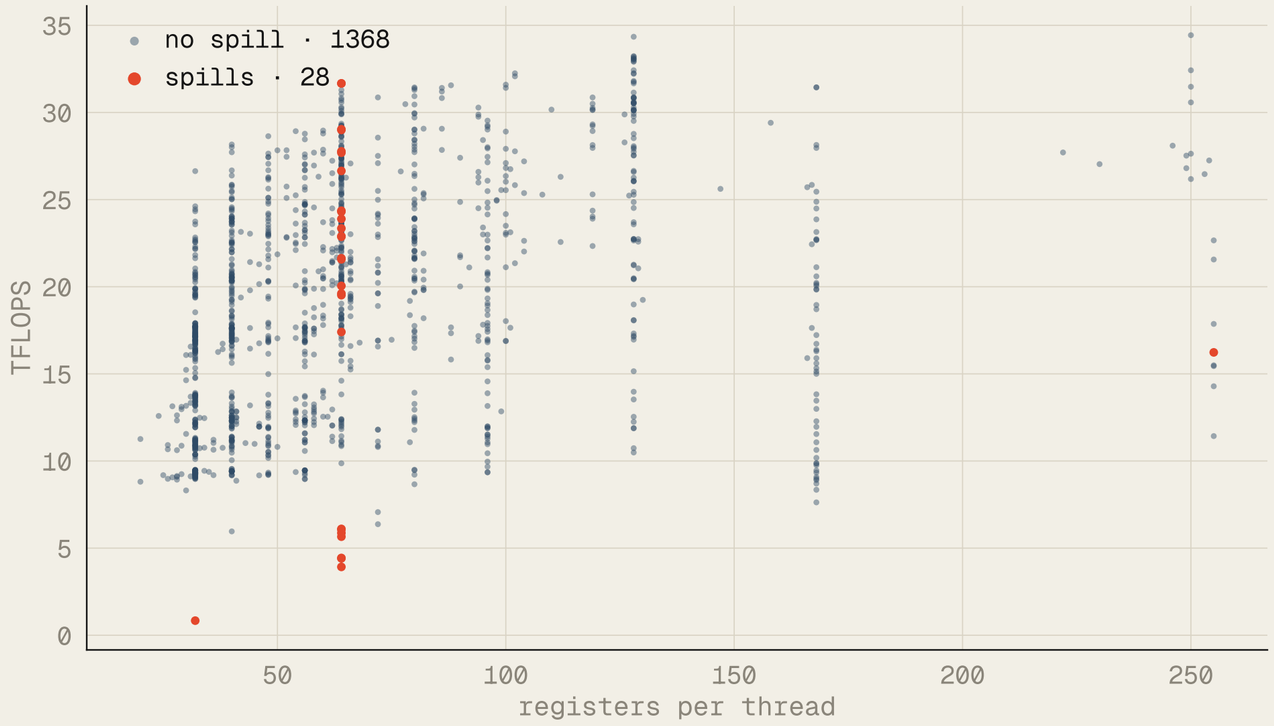


FIG. 6 — REGISTERS PER THREAD VS THROUGHPUT, ALL 1,396 CONFIGURATIONS

SPILLING

28

of 1,396 configurations spill registers at all

MEDIAN TFLOPS

21.6 vs 19.0

spilling vs not: spillers are not slower

RANK CORRELATION

ρ +0.47

more registers, more speed: tile reuse beats occupancy

What the results do not show

01

One deviation

After the gate failed, the models only proposed candidates; measurement chose.

02

Winner from screening

The selected configuration is a corner of the screening design; later stages confirmed it.

03

Optimum out of reach

The true best (BK = 64) lies outside the region the models were allowed to search.

04

Narrow scope

One GPU, one kernel family, FP32 on CUDA cores, tuned at 4096^3 only.

05

Shared, clamped GPU

Other users' jobs were detected and re-run; the clock held at 1830, not 1980 MHz.

06

Discrete lattice

Power-of-two levels forced design points to snap; the CCD centre is not the true centre.

Few experiments, a near-best kernel, *an honest model*

RQ1	SCREENING	Yes: all six factors separated (R^2 0.959); importance shifts at 1024^3
RQ2	QUADRATIC	Close but inadequate: lack of fit 7.8% against a 5% limit
RQ3	EFFICIENCY	Yes: 59 evaluations against a median of 424 for random search
RQ4	OPTIMALITY	0.17% from the exhaustive optimum, #2 of 1,396
RQ5	CUBLAS	72.0% of FP32 throughput; 2.6× the untuned kernel

Use the response surface to propose; let measurement decide.

Where this goes next

01

Larger design

A full 2^6 core to estimate 3-way interactions without overfitting

02

Per-shape tuning

Importance shifts below 2048^3, so tune each matrix size

03

Stronger baseline

Head-to-head with Bayesian optimisation, as modern autotuners use

04

Richer kernel

Vectorised loads and double buffering as categorical factors

Thank you.

QUESTIONS WELCOME